Department of Computer Science

Project Descriptions Summary

COS40005 Computing Technology Project A
Allocated projects, September 2026 intake

Projects	11 allocated projects across 12 teams
Industry	Huawei, AWS FCAJ (VIB, Katalon, Cloud Kinetics, Quỹ Nuôi Em), ISC2 Vietnam Chapter
Research	AI02: Vietnamese Sign-Language Recognition
Self-sourced	Multimodal Deepfake Detection; AI Automation for Counting Milk (Datagent)
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01 Overview

This document describes every project allocated for Project A. Read your own project carefully before your first client meeting. Each industry project lists the options you can bring to that meeting; your client may also propose an alternative, and that is equally valid.

#	Project	Client / mentor	Team
2.1	AI FitWear: Smart Fitness & Watch Ecosystem	Huawei · Linh Ngô, Khy Hoong	Group 1
3.1	VIB 1: Internal Assistant for Enterprise Knowledge	AWS FCAJ · Phạm Tiến Thuận Phát	Group 3
3.2	VIB 2: Consistent AI-Generated Output for Internal Documents	AWS FCAJ · Aiden Dinh	Group 2
3.3	Customer Support Email Campaign Platform	AWS FCAJ · Lê Minh Nghĩa	Group 8
3.4	Natural Language Query Agent for Enterprise Data	AWS FCAJ · Khang Nguyen	Group 4
3.5	Quỹ Nuôi Em: Social-Enterprise Technology Topics	AWS FCAJ · Trần Hữu Nghĩa	Group 5
4.1	ISC2 Topic 3: Security Validation for Agentic AI Systems	ISC2 · Robert Leyba	Groups 7, 10
4.2	ISC2 Topic 4: Human-AI Malware Triage and Threat Analysis	ISC2 · Robert Leyba	Group 9
5.1	AI02: Vietnamese Sign-Language Recognition	Assoc. Prof. Dr. Dang Tran Khanh	Group 11
6.1	Multimodal Deepfake Detection	Dr. Nguyen Xuan Sam	Group 6
6.2	AI Automation for Counting Milk	Datagent · Peter Lê	Group 12

Before your first client meeting

- Book the meeting as soon as possible, ideally within the next few days, copying danieltnguyen@swin.edu.au
- Read your project section as a team and shortlist the options you want to pursue
- Prepare questions on scope, data and platform access, and what a realistic Project A deliverable looks like
- Agree who leads the conversation and who takes notes, and settle a communication channel and meeting rhythm with your client

02 Huawei

2.1 AI FitWear: Smart Fitness & Watch Ecosystem Platform

Huawei CBG · Mentors: Linh Ngô (BD Manager, Huawei Vietnam), Khy Hoong (Software Technical Support, Huawei Cloud APAC)
Allocated: Group 1 (Hồ Quốc Khánh)

A closed-loop fitness system for beginner gym-goers that removes manual food logging and provides in-workout guidance without a personal trainer. A mobile app acts as the "central brain" (AI meal-photo analysis, BMR/TDEE calculation, six-muscle-group programme generation, tutorial videos), while a HarmonyOS watch app acts as the sensor agent (IMU-based rep counting, haptic rest timer, calorie-deficit dashboard). The two devices stay synchronised in real time.

Scope options to discuss with your mentors
1. Which module to build first in Project A: meal-photo AI, watch rep counting, or the sync engine
2. Meal recognition via a hosted multimodal API or a custom fine-tuned model
3. Target platforms for the mobile app (iOS, Android or cross-platform)
Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Mobile app (iOS/Android) with AI food-image recognition, macro breakdown, body metrics and automated workout planner
- HarmonyOS watch app with wrist dashboard, micro-animation form previews, IMU rep counting (8–12 reps) and haptic rest alerts
- Bi-directional, low-latency sync engine (Bluetooth LE or WebSockets) between phone and watch
- Documentation and validation report: architecture, meal-recognition accuracy, motion-detection precision, user testing

Technology

ArkTS in Huawei DevEco Studio for the watch app; mobile framework at the team's discretion; multimodal vision API (GPT-4o Vision, Gemini Flash or similar) or a fine-tuned YOLO/MobileNet model; Firebase or AWS Amplify, Node.js or Python FastAPI backend.

03 AWS FCAJ

AWS FCAJ coordinates projects from VIB, Katalon, Cloud Kinetics and Quỹ Nuôi Em. Evaluation weights marked "suggested" were proposed by Swinburne and are pending FCAJ confirmation.

3.1 VIB 1: Internal Assistant for Enterprise Knowledge

AWS FCAJ (VIB use case) · Mentor: Phạm Tiến Thuận Phát, Cloud Solutions Architect, VIB
Allocated: Group 3 (Trương Ngọc Gia Hiếu)

Company knowledge is scattered across Confluence spaces (Products, Legal, Policies, Benefits), a documentation site and forums, with no single way to query it. The team builds a Slack chatbot that gives employees accurate answers drawn from these sources.

Options to discuss with your client
1. Retrieval approach: MCP server or Bedrock Agent
2. Which knowledge sources to ingest first (Confluence, documentation site, forums)
3. How strictly to model document-level permissions in Project A

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Slack-integrated chatbot answering from the internal knowledge base
- Ingestion pipeline that keeps the knowledge base current as documents change
- Access-control layer so answers draw only on documents the employee may read

Technology

Python or TypeScript; Amazon Bedrock; Slack API with Confluence and web connectors; MCP server or Bedrock Agent for RAG.

Evaluation

Suggested: answer accuracy and relevance 40%, knowledge-base freshness 30%, access-control correctness 30%.

3.2 VIB 2: Consistent AI-Generated Output for Internal Documents

AWS FCAJ (VIB use case) · Mentor: Aiden Dinh, Operations Engineer, Katalon

Allocated: Group 2 (Nguyễn Hữu Thiên Khang)

Employees draft reports and BRDs with a mix of individually licensed AI tools, and the drafts are not checked against internal rules, policies or templates, so downstream teams rework them. The team designs a way for employees to produce compliant, on-template output as part of normal day-to-day AI use.

Solution options from the brief

1. A shared, policy-aware prompt library
2. A template-aware AI assistant
3. A lightweight review layer that checks AI output before it is shared

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- A mechanism that applies internal rules and templates to AI-generated reports and BRDs
- A low-friction way to use it inside existing daily workflows
- Demonstrable reduction in rework caused by non-compliant AI output

Technology

Amazon Bedrock or the organisation's Claude Team subscription; MCP server or a policy-aware prompting layer; Python or TypeScript. (FCAJ did not specify a stack; this is proposed.)

Evaluation

Suggested: ease of adoption 40%, output compliance 40%, measured rework reduction 20%.

3.3 Katalon: Customer Support Email Campaign Platform

AWS FCAJ (Katalon assignment) · Mentor: Lê Minh Nghĩa, Cloud Engineer, ITRVN

Allocated: Group 8 (Cao Việt Anh)

Katalon's support teams need to run email campaigns, such as appointment reminders, reliably and at scale. The team builds a campaign platform on Amazon Connect and Amazon SES with dynamic templates, segmentation, scheduling and resilient bulk sending.

Options to discuss with your client

1. Queue technology: SQS, RabbitMQ or Kafka
2. How far into the advanced features (dashboard, monitoring, audit logs) to go in Project A
3. Whether to add a custom application layer and Infrastructure as Code from the start

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Campaign creation with dynamic placeholder templates ({{first_name}}, {{company}})
- Segment-based, scheduled bulk delivery with retry for transient failures
- Campaign history, email logs and a dashboard (recipients, success/failure, delivery rate, optional open/click rate)
- Monitoring, failed-campaign alerts and audit logs

Technology

Amazon Connect and SES; queue-based architecture (SQS, RabbitMQ or Kafka) with workers; idempotent, secure template rendering; Infrastructure as Code is a plus.

Evaluation

Suggested: core features 50%, advanced features 30%, architecture and best practice 20%.

3.4 Cloud Kinetics: Natural Language Query Agent for Enterprise Data

AWS FCAJ (Cloud Kinetics) · Mentor: Khang Nguyen

Allocated: Group 4 (Lê Thành Nam)

A large real-estate corporation holds around 100 raw database tables across development, sales, finance, procurement, construction and property management, and business users cannot query them without technical help. The team researches and builds an AI agent that answers business questions in natural language, grounded in the data, and compares alternative approaches rather than assuming one pattern. Synthetic or anonymised data only.

Example approaches to compare (the brief leaves the method open)

1. Text-to-SQL with schema retrieval that selects the relevant tables per question
2. A semantic layer or data catalogue that documents table meaning before querying
3. A multi-step agent that plans, queries, checks and refines its answer

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Formal proposal: problem, objectives, approach, assumptions, work plan, risks, evaluation method
- Justified solution architecture covering data flow, security and integration points
- Defined scope: dataset, supported questions, constraints and out-of-scope items
- Working AI agent over a ~100-table environment, including cross-domain questions
- Hosting environment, plus monitoring of performance, failures, AI interactions and data access
- Technical documentation, evaluation results and recommendations

Technology

Not prescribed; students select and justify the stack on technical, security, operational and cost grounds.

Evaluation

Accuracy, scalability, maintainability and usability, measured on a representative set of business questions.

3.5 Quỹ Nuôi Em: Social-Enterprise Technology Topics

AWS FCAJ (Quỹ Nuôi Em, via VIB) · Mentor: Trần Hữu Nghĩa, Cloud Solutions Architect, VIB

Allocated: Group 5 (Phạm Thành Hưng); specific topic to be agreed at the kickoff meeting

Quỹ Nuôi Em runs a "one NE code, one student, one sponsor" model funding lunches for ethnic-minority students, alongside Sức Mạnh 2000 (schools, bridges, houses), Phòng Tin Học Cho Em and an industrial-kitchen programme. Large data volumes, transparency needs and volunteer-heavy operations create demand for standardisation and automation. The fund has proposed the ten topic options below.

Mandatory principles for every topic: never publish children's data beyond fund-approved fields; role-based access for photos, names, schools, contacts and transactions; AI only suggests, humans approve anything touching child profiles or reconciliation; audit logs for financial and code-matching changes; Excel import/export kept during transition; public dashboards show aggregated, approved data only; mobile-first.

#	Topic	Type	What the team would build
1*	**AI Student Photo Processing & Profile Card Generator**	AI/ML, computer vision	Batch tool that detects and crops faces, scores photo quality, applies the Nuôi Em frame and exports by student code, with a manual review queue.
2*	**"Hỏi Nuôi Em" RAG Chatbot for Sponsors**	GenAI, RAG	Website chatbot answering sponsor FAQs from approved sources with citations, admin document upload and handoff to human support.
3	**AI Transaction Reconciliation Assistant**	AI/ML, FinTech	Bank-statement parser with fuzzy matching of transfers to sponsor codes, review statuses, audit log and Excel export.
4*	**Nuôi Em Core: Sponsor–Student–Transaction System**	Software, database	Normalised data model and internal MVP covering the student, code, sponsor, transaction and disbursement lifecycle, with RBAC and Excel import.
5*	**Financial Transparency & School Disbursement Tracker**	FinTech, data platform	Cash-flow and disbursement tracking per project and school with internal approval and a public aggregated dashboard.
6	**Bulk Statement Lookup Portal for Sponsors**	Software, self-service	Secure lookup by code plus minimal verification, returning safe results instead of full statements.
7	**Redesign 3 Websites with a Shared Design System**	Web design, frontend, UX	Responsive redesign with a shared design system across the fund's sites.
8	**Impact Story & Project Progress Map**	Web, data viz, GIS	Public portal mapping projects with progress, photos, funding needed and completion reports.
9*	**Donation Platform for Sức Mạnh 2000**	Full-stack, payments	Donation MVP with PayOS/Casso integration if sandbox access is granted, plus bank transfer support.
10	**School Needs Assessment & Prioritisation Platform**	Data, decision support	Intake and explainable scoring of school requests to produce a prioritised shortlist for review.

* The fund's top five priorities. Please shortlist two or three topics as a team before the meeting. *Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.*

04 ISC2 Vietnam Chapter

ISC2 topics are research directions rather than fixed specifications. Teams refine the research question with Robert Leyba (Vice President, ISC2 Vietnam Chapter) and their supervisor, then build a prototype or testbed, establish a baseline, run repeatable experiments and report failure cases as well as successes. All testing stays in authorised, isolated lab environments with synthetic or sanitised data.

4.1 Topic 3: Security Validation for Agentic AI and Tool-Using Systems

ISC2 Vietnam Chapter · Mentor: Robert Leyba

Allocated: Group 7 (Đoàn Gia Hân) and Group 10 (Trần Phương Hà)

Research question: how can security controls limit harmful or unauthorised actions when an AI agent can access tools, external data, APIs or autonomous workflows? Teams build a lab agent with simulated tools, define trust boundaries, run safe adversarial scenarios and validate whether controls prevent or contain unauthorised behaviour.

Possible focus areas (Groups 7 and 10 should agree different focus areas with Robert)

1. Indirect prompt injection through manipulated external content
2. Unauthorised or unsafe tool use and privilege misuse
3. Unsafe chaining of actions across multiple tools
4. Effectiveness of controls: least-privilege scopes, allowlists, human approvals, audit logging

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Agentic-AI lab environment with a trust-boundary and tool-permission model
- Threat model and repeatable security test scenarios
- Agent action and audit logging pipeline
- Security-control validation matrix with before-and-after evaluation
- Safe-deployment and human-approval recommendations; final research report and demo

Evaluation

Unauthorised-action success, control prevention rate, approval effectiveness, audit completeness, privilege exposure, task-success trade-off.

4.2 Topic 4: Human-AI Malware Triage and Threat Analysis

ISC2 Vietnam Chapter · Mentor: Robert Leyba

Allocated: Group 9 (Đỗ Đăng Khoa)

Research question: can AI-assisted malware triage reduce analyst effort while keeping findings reliable, explainable and evidence-supported? Teams build a safe pipeline that turns lab analysis artefacts into structured evidence, add an AI triage layer, and compare analyst, AI and analyst-plus-AI outputs against a benchmark. Malware work stays in isolated labs using approved or benign samples.

Possible focus areas

1. AI summarisation of static and behavioural evidence
2. Automated IOC extraction and MITRE ATT&CK; mapping
3. Analyst-only vs. AI-only vs. analyst-plus-AI comparison on a labelled benchmark
4. Measuring unsupported claims and missed evidence in AI output

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Key deliverables

- Safe malware-analysis and triage environment with a structured evidence pipeline
- AI-assisted triage prototype
- Benchmark dataset with documented ground truth
- Human vs. AI vs. human-plus-AI evaluation, with error and hallucination analysis
- Analyst safe-use guidance; final research report and demo

Evaluation

Triage time, behavioural accuracy, IOC and ATT&CK; mapping accuracy, unsupported-claim rate, analyst-effort reduction.

05 Research track

5.1 AI02: Vietnamese Sign-Language Recognition for a Defined Vocabulary

Research track · Supervisor: Assoc. Prof. Dr. Dang Tran Khanh
Allocated: Group 11 (Huỳnh Gia Phúc)

A camera application that recognises a limited vocabulary of isolated Vietnamese signs, using consented recordings from multiple signers, landmark extraction and a deep-learning sequence classifier. As with all research topics, the AI component must be compared against a simple baseline; Project A targets a scoped prototype with initial validation.

Key deliverables

- Defined vocabulary and signer-independent data splits with a simple baseline
- Working camera prototype with latency measurement
- Extension: robustness to lighting and background changes, deployment with signer feedback

Evaluation

Macro-F1 on unseen signers, latency and per-sign confusion.

06 Self-sourced projects

6.1 Multimodal Deepfake Detection with Audio-Visual and Physiological Cues

Self-sourced research · Swinburne Vietnam · Mentor: Dr. Nguyen Xuan Sam (sxnguyen@swin.edu.au)
Allocated: Group 6 (Vũ Bình Phương)

Deepfake video calls forging both face and voice are increasingly used in scams, and current detectors degrade under call compression or generalise poorly to new generators. The team tests whether breathing is consistent across audio (audible inhalations) and video (shoulder and chest motion, respiratory modulation of the facial pulse) in genuine speakers, and fuses this cross-modal respiratory signal with existing audio-visual detectors. The detector returns a risk score with a human-readable reason and abstains when the signal is too weak.

Key deliverables

- Feasibility studies on audio-video breathing coupling and codec robustness
- Preprocessing pipeline with rPPG extraction
- Cross-modal consistency module with baseline comparison
- File-scan prototype returning a risk score and explanation
- Sprint reports, video presentation, research paper submission, final documentation and source code

Technology

Python, PyTorch, scikit-learn; OpenCV and MediaPipe; torchaudio or librosa; FFmpeg for call-codec simulation; Git/GitHub; public audio-visual deepfake and rPPG datasets only.

6.2 AI Automation for Counting Milk

Self-sourced industry project · Datagent · Mentor: Peter Lê (phle@swin.edu.au)

Allocated: Group 12 (Nguyễn Huy Hoàng)

The team will train a model that recognises milk products from cameras in grocery stores, so that product counts can be collected automatically. The full project brief is held by the team; scope, dataset and evaluation measures are to be confirmed with the mentor at the kickoff meeting.

Scope questions to settle with your mentor

1. Recognition scope: detect milk products only, or identify brand and SKU
2. Data source: in-store camera footage, collected photos or public retail datasets
3. Output: live counts, periodic shelf snapshots, or a reporting dashboard

Your client may also propose an alternative option. Treat it as a valid choice and discuss it openly.

Technology

AWS; Python backend; Next.js frontend.